

PAPER_863: Water Reactor Birkeland-Current H₂/O₂ Electrolysis Efficiency

****Author:**** Daniel T. Murphy -- Star Magic / UQFF Framework

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****Session:**** 200

****Source:**** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)

****Calculator:**** WaterReactorBirkelandH2ElectrolysisEfficiencyCalc (CP4 #447)

****CVW:**** v2.0.0 compliant

Abstract

We present a UQFF-framework calculator for a water reactor system exhibiting 283:1 energy efficiency via H₂/O₂ electrolysis with Birkeland current banding. The system uses 27 W input to process 75.7 L/min water flow producing 107 L/min H-O gas (H₂: 71.34 L/min = 0.0531 mol/s, O₂: 35.66 L/min = 0.0265 mol/s). A 237 mL/h surplus water generation from atmospheric condensation and a 100-ft (30.5 m) repellent field complete the energy budget: E_{input} = 194,400 J yields E_{out} = 55,069,803 J over 2 hours.

1. Core Equations

- $E_{input} = P * t$ (P = 27 W, t = 7200 s)
- $H2_{mol_rate} = V_{gas} * 0.667 / 22.4 / 60$ (STP molar volume)
- $E_{gas} = H2_{mol_rate} * 286000 * t$ (H₂ combustion enthalpy 286 kJ/mol)
- $E_{surplus} = surplus_mass * 2257$ (latent heat of water)
 $eta = E_{total} / E_{input} = 283:1$
- $J_{Birk} = 1e-5 * (V_{gas} / V_{flow})$ (Birkeland current density heuristic)

2. UQFF Integration

Birkeland current banding is the laboratory-scale analog of Ug3 magnetic string-disk topology. The surplus water condensation from atmospheric coupling maps to Aether-mediated energy exchange. This calculator operates as a stateless physics calculator within CondensedPhysics4.py. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline.

3. Source Data

- ****File:**** advanced_system_analysis_simulator_quantum_calculator.txt (3745 lines)
- ****Session:**** 200
- ****Experimental specs:**** 27 W input, 75.7 L/min water, 107 L/min gas, 237 mL/h surplus, 30.5 m field
- ****VDS/DVP/BH:**** ABSENT

References

1. Murphy, D.T. -- Star Magic UQFF Framework (2024-2026)
2. Birkeland, K. -- The Norwegian Aurora Polaris Expedition 1902-1903 (1908)
3. NIST Standard Reference Database -- H°_f combustion enthalpy, latent heat values
4. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $\beta_i \sim 0.603$